PROJECT PROPOSAL

Email Marketing Management System

AI-Powered Web-Based Platform

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A practical application of web development, AI-assisted content creation, email campaign management, and performance tracking.

1. Abstract

This project proposes an AI-Powered Email Marketing Management System, a web-based platform that helps businesses create, personalize, schedule, and evaluate email marketing campaigns. It will provide a simple dashboard where users can manage subscribers, prepare campaign content, schedule emails, and review campaign reports. An AI API will help users draft marketing emails, newsletters, and subject lines in English and Urdu, while users will review and approve the generated content before sending. To make the system more distinctive than a basic email campaign manager, it will include a simple email warm-up tool, inbox rotation for configured sending accounts, and A/B testing for comparing campaign variations. The project will be evaluated through functional testing, usability checks, and campaign metrics supported by the selected email delivery service.

2. Introduction

Email marketing is an important way for businesses and organizations to share announcements, promote products, introduce services, and maintain communication with customers. It can help organizations reach subscribers directly and provide information about new offers, events, and updates. However, preparing suitable email content, maintaining contact lists, planning campaign delivery, and checking results can require considerable manual effort.

This project aims to bring these tasks together in one web-based system. Artificial Intelligence will assist users with writing and improving email content, while campaign tools will help organize contacts, schedule messages, and review results. Additional features such as warm-up support, inbox rotation, and A/B testing will make the project more focused on practical email campaign workflows.

3. Problem Statement

Small businesses and organizations may not have access to an affordable and easy-to-use platform for managing their email marketing activities. When different tasks are performed manually or through separate tools, users may find it difficult to keep subscriber records organized, prepare consistent campaign content, schedule emails efficiently, and compare campaign performance.

The proposed system addresses these challenges by providing a centralized dashboard for subscriber management, AI-assisted writing, campaign scheduling, and reporting. It will also include a basic warm-up workflow, inbox rotation for approved sending accounts, and A/B testing so users can compare selected email variations and make decisions based on available results.

4. Aim and Objectives

The main aim is to develop a practical web-based system that simplifies email marketing management and supports more efficient campaign creation, delivery, and evaluation.

- Provide a central dashboard for managing subscriber records and email campaigns.
- Use AI to suggest email drafts, newsletters, and subject lines in English and Urdu.
- Allow users to create, edit, preview, and schedule email campaigns.
- Provide a basic warm-up workflow that supports gradual test-email sending within configured limits.
- Rotate campaign sending between connected and authorized inboxes when supported by the selected email service.
- Compare two subject lines or email versions through A/B testing and summarize the available results.
- Protect subscriber preferences by recording unsubscribe requests and excluding opted-out contacts from future marketing campaigns.

5. Project Scope

The system will provide a dashboard for authorized users to manage subscribers and email campaigns. Users will be able to add, edit, and organize contact records; create email templates; generate AI suggestions; group subscribers; schedule campaigns; and review available reports. The interface will be designed to keep the main tasks easy to understand, even for users with limited technical experience.

The email warm-up feature will provide a basic process for gradually sending test messages within selected limits. Inbox rotation will distribute eligible campaign messages among connected and authorized sending inboxes, subject to the selected provider's rules, permissions, and sending limits. A/B testing will let users compare two selected versions, such as different subject lines or email content, and review metrics provided by the email service.

The project will focus on a manageable prototype suitable for small businesses and organizations rather than a full commercial email deliverability platform. Open and click tracking will be implemented only where supported by the selected delivery service. The system will also respect subscriber consent, provide unsubscribe handling, and avoid sending marketing messages to contacts who have opted out.

6. Proposed Methodology

Phase 1 — Requirements and Design: Identify user needs, define the main features, plan the database tables, and prepare simple interface sketches.

Phase 2 — Core Application: Build login, the dashboard, subscriber management, campaign creation, and the basic database operations.

Phase 3 — Email Integration: Connect an SMTP service or email delivery API and add campaign scheduling and sending status records.

Phase 4 — AI and Distinctive Features: Connect an AI API for email drafts and subject lines in English and Urdu, then develop a basic warm-up workflow and inbox rotation for configured inboxes.

Phase 5 — Testing and Analytics: Add A/B testing, display available delivery and engagement results, and test the system's main workflows, security, and usability.

Phase 6 — Deployment and Documentation: Deploy the prototype where resources permit, correct issues found during testing, and prepare technical documentation and the final report.

7. Proposed System Architecture

The system will use a simple web frontend built with HTML, CSS, and JavaScript, connected to a Python Django backend. The frontend will provide forms and dashboard pages, while the backend will process requests, apply business rules, and communicate with the database and external services.

A database such as SQLite for the initial prototype or PostgreSQL for a larger setup will store user accounts, subscriber records, campaigns, templates, sending settings, and available campaign statistics. The backend will connect to an AI API for content suggestions and to an SMTP service or email delivery API for sending messages. Basic scheduling can be handled through a simple scheduled task or the selected provider's scheduling features. Chart.js may be used to present campaign statistics in a clear dashboard.

8. Tools and Technologies

The project will use technologies that are suitable for a student-level prototype and can be developed in stages. The final selection may depend on available hosting resources and the email provider's integration options.

Component	Proposed Technology
Frontend	HTML, CSS, and JavaScript
Backend	Python and Django
Database	SQLite for the prototype; PostgreSQL if needed
AI Integration	LLM API with simple prompts
Email Delivery	SMTP or a suitable email delivery API
Scheduling	Basic scheduled tasks or provider scheduling
Charts	Chart.js, if required
Development Tools	Visual Studio Code and GitHub

9. Main Features

- **User Authentication:** Secure login will allow authorized users to access the campaign dashboard and manage their own campaign information.
- **Subscriber Management:** Users will be able to add, edit, import, and organize subscriber records. Basic validation will help reduce incomplete or incorrectly formatted contact details.
- **AI Email Generator:** The system will generate draft emails, newsletters, promotional messages, and subject-line suggestions from user instructions. Users will be able to review and edit drafts before use.
- **English and Urdu Content:** AI-assisted content generation will support both languages, depending on the selected model's quality. Users will be responsible for checking the final wording.
- **Campaign Management:** Users will be able to create, edit, preview, and manage campaigns before sending them to selected subscriber groups.
- **Scheduling:** Users will be able to set a planned delivery time through the configured email service or the system's basic scheduling feature.
- **Email Warm-Up Tool:** A basic, controlled workflow will support gradual test-email sending within configured limits. It will record relevant activity where possible, but it will not guarantee inbox placement.
- **Inbox Rotation:** Where supported, sending can rotate among connected and authorized inboxes to distribute eligible campaign traffic within provider rules and account limits.
- **A/B Testing:** Users can compare two subject lines or email versions and review available delivery, open, or click metrics to see which version performs better. Conclusions will depend on actual results and the number of recipients.
- **Analytics and Reports:** The dashboard will show delivery status, bounces, opens, and clicks when the selected provider makes these metrics available.
- **Unsubscribe Management:** The system will record opt-outs and exclude unsubscribed contacts from future marketing campaigns to respect subscriber preferences.

10. Functional Requirements

The functional requirements describe the main tasks that the system is expected to perform.

- The system shall allow authorized users to sign in and access campaign management features.
- The system shall allow users to add, update, import, and organize subscriber records.
- The system shall allow users to create, edit, preview, and save email campaigns and templates.
- The system shall connect to a configured AI API to generate draft email content and subject-line suggestions.
- The system shall support sending or scheduling campaigns through a configured email delivery service.
- The system shall provide basic warm-up settings and keep test sending within configured limits.
- The system shall rotate sending across connected, authorized inboxes only where the provider and account setup permit it.
- The system shall allow users to compare two campaign variants and display available comparison metrics.
- The system shall store available campaign delivery results and display them in reports.
- The system shall process unsubscribe requests and prevent opted-out contacts from receiving future marketing campaigns.

11. Non-Functional Requirements

- **Usability:** The interface should be clear, consistent, and easy to navigate.
- **Security:** Login protection, access controls, and safe handling of API credentials should be used.
- **Reliability:** The system should validate user inputs and record sending or scheduling errors where possible.
- **Maintainability:** The code should be organized into clear frontend, backend, and email-service parts so that changes are easier to manage.
- **Privacy:** Subscriber information should be handled responsibly, and consent and unsubscribe preferences should be respected.
- **Compatibility:** The main dashboard should work in a modern web browser on common desktop and laptop screens.

12. Expected Outcomes

The expected outcome is a working web application prototype that brings subscriber management, AI-assisted email writing, campaign scheduling, a basic warm-up workflow, inbox rotation, A/B testing, and campaign reports into one platform. Users should be able to follow a clear workflow: organize subscribers, prepare an email, review AI suggestions, select a campaign audience, schedule or send the campaign, and examine the available results.

The project will demonstrate how web development, database operations, AI integration, and email delivery services can work together in a practical application. The final results will depend on the capabilities of the selected services and the outcomes of testing.

13. Evaluation and Testing

Testing will cover login, subscriber management, AI content generation, campaign creation, scheduling, warm-up limits, inbox rotation, A/B testing, unsubscribe handling, and report accuracy. Test cases will be prepared to check both successful actions and common errors, such as invalid subscriber details, missing campaign information, or unavailable email-service connections.

The project will be assessed using completed test cases, user feedback where available, processing and sending records, and supported engagement metrics. A/B testing will be checked using actual campaign variations and available results. Conclusions about which version performs better will be based on observed data and sufficient observations rather than assumed in advance.

14. Project Timeline

Period	Planned Activities
Month 1	Requirements gathering, system design, database planning, and initial interface.
Month 2	Login, subscriber management, campaign creation, and email service connection.
Month 3	AI writing, warm-up workflow, inbox rotation, A/B testing, and basic reports.
Month 4	Testing, improvements, deployment where possible, documentation, and final presentation.

15. Limitations and Future Enhancements

Email delivery, open tracking, and click tracking depend on the selected provider's capabilities and policies. Some providers may restrict sending volume or the number of connected inboxes. Warm-up and inbox rotation will therefore depend on provider rules, account permissions, and sending limits; neither feature can guarantee inbox placement or successful delivery to every recipient.

AI-generated content may require human review for accuracy, tone, language quality, and suitability for the intended audience. A/B testing is most useful when enough recipients and reliable engagement data are available. For a small prototype, the results may be limited, and advanced predictions will not be treated as a core requirement.

Future enhancements may include improved subscriber segmentation, send-time recommendations, integration with online stores, reusable campaign templates, and more detailed campaign insights if sufficient data and resources become available.

16. Conclusion

The AI-Powered Email Marketing Management System will provide a centralized and practical way to manage subscribers and email campaigns while using AI to assist with content creation. Its basic warm-up workflow, inbox rotation, and A/B testing will give the project more specific goals than a general email campaign manager. The use of a simpler technology stack will keep development manageable while still demonstrating web development, database management, AI integration, email automation, and software testing. The proposed system is intended to help users organize campaign tasks, reduce repetitive work, and review performance using the information available from the selected email service.